

SMART IRRIGATION SYSTEM

Team No: 07

TEAM INFORMATION:

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PROJECT OBJECTIVE:

- ▶ The ESP32-powered smart irrigation system aims to provide an efficient, automated, and sustainable way of managing and utilizing water in agricultural fields, gardens, or other landscapes. Its main goal should be the optimization of irrigations with real-time monitoring of soil moisture, together with environmental conditions while keeping the loss of excess water as low as possible. As a result, healthy plants would be produced. It aims to integrate IoT technologies for remote monitoring and control, convenience for users, and reduced manual intervention towards resource conservation and sustainable farming practices.

COMPONENTS USED:

▶ •HARDWARE:

- ▶ 1. ESP 32 MICROCONTROLLER
- ▶ 2. SOIL MOISTURE SENSOR
- ▶ 3. LED SCREEN
- ▶ 4. BATTERY
- ▶ 5. ACTUATOR
- ▶ 6. POWER SUPPLY
- ▶ 7. WIFI MODULAR
- ▶ 8. BREADBOARD
- ▶ 9. JUMPER WIRES

SOFTWARE:

- 1.THONNY
- 2.BLINK

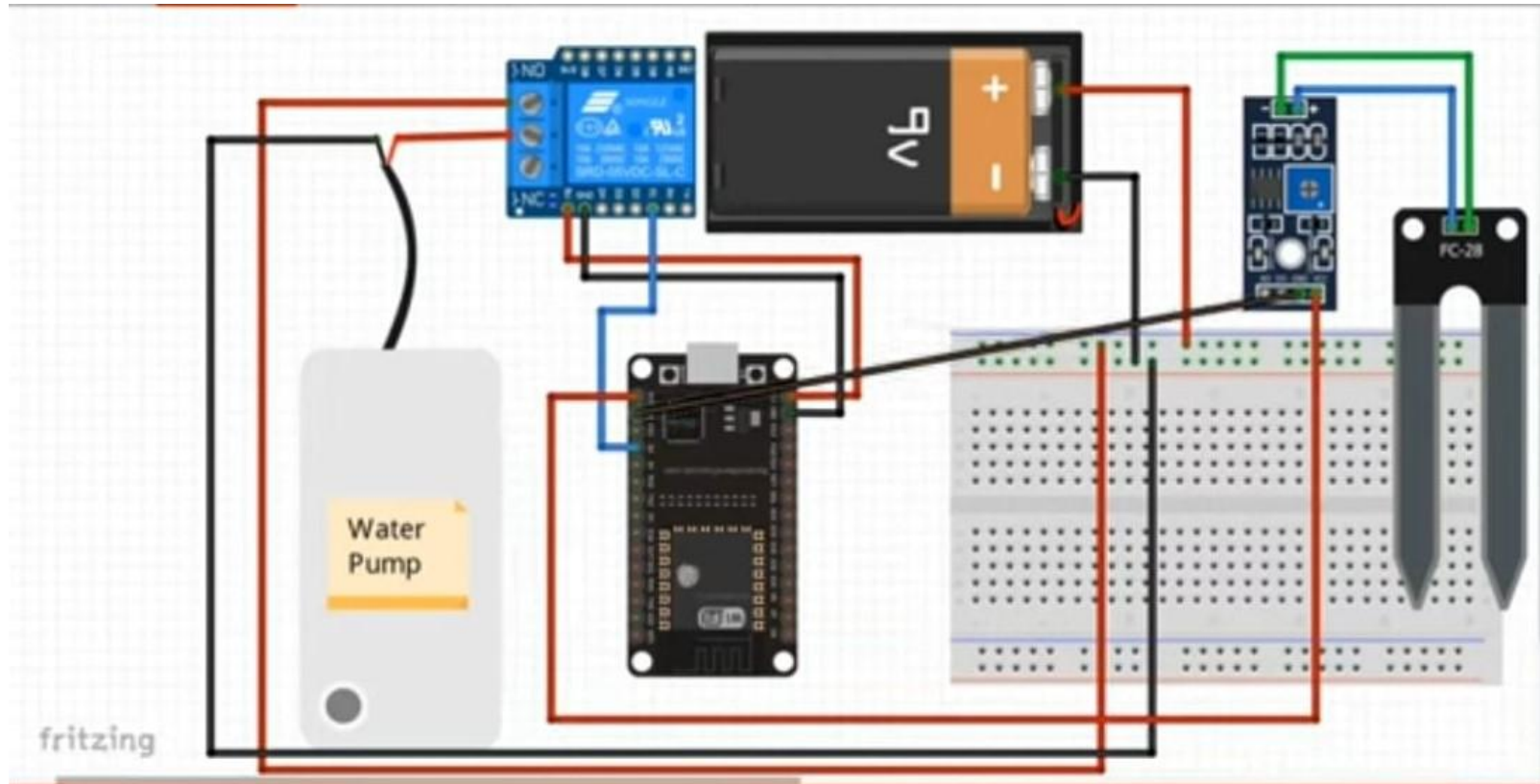
LITERATURE SURVEY:

Sl.no	Paper title	Author	Key contribution	Advantages	Disadvantages	Tools
01	Design and Development of IoT-Based Smart Irrigation System	Dr. Vivek Vaidya, Mr. Yash Chavan, Miss. Dineshwari Madavi, Mr. Utkarsh Meshram, Mr. Somesh Nandeshwar	This literature highlights the importance of smart irrigation systems in addressing the global water crisis, emphasizing their role in conserving water and optimizing agricultural practices.	Optimize water usage and prevent water wastage with precise irrigation control	High initial setup cost	Soil moisture sensor, Microcontroller
02	IoT-Based Approach for Smart Irrigation System Suited to Multiple Crop Cultivation	Krishna Singh, Samyak Jain, Varun Andhra, Shilpi Sharma	The paper proposes an advanced automated irrigation system tailored to India's multi-crop agriculture, integrating IoT and data science technologies for efficient crop cultivation.	Ensure healthy plant growth by maintaining soil moisture levels	Not suitable for all types of crops	Water controller, Power supply
03	An Overview of Smart Irrigation Systems Using IoT	Khaled Obaideena, Bashria A.A. Yousef, Maryam Nooman AlMallahi, Yong Chai Tan, Montaser Mahmouda, Hadi Jaber, Mohamad Ramadan	This literature review explores the role of smart irrigation in advancing Sustainable Development Goals, with a focus on water-use efficiency through IoT and sensory systems.	Enjoy convenience of remote monitoring and control via the blink app	Installing and maintaining the equipments	Actuators, Power supply, Soil moisture sensor/flow sensor

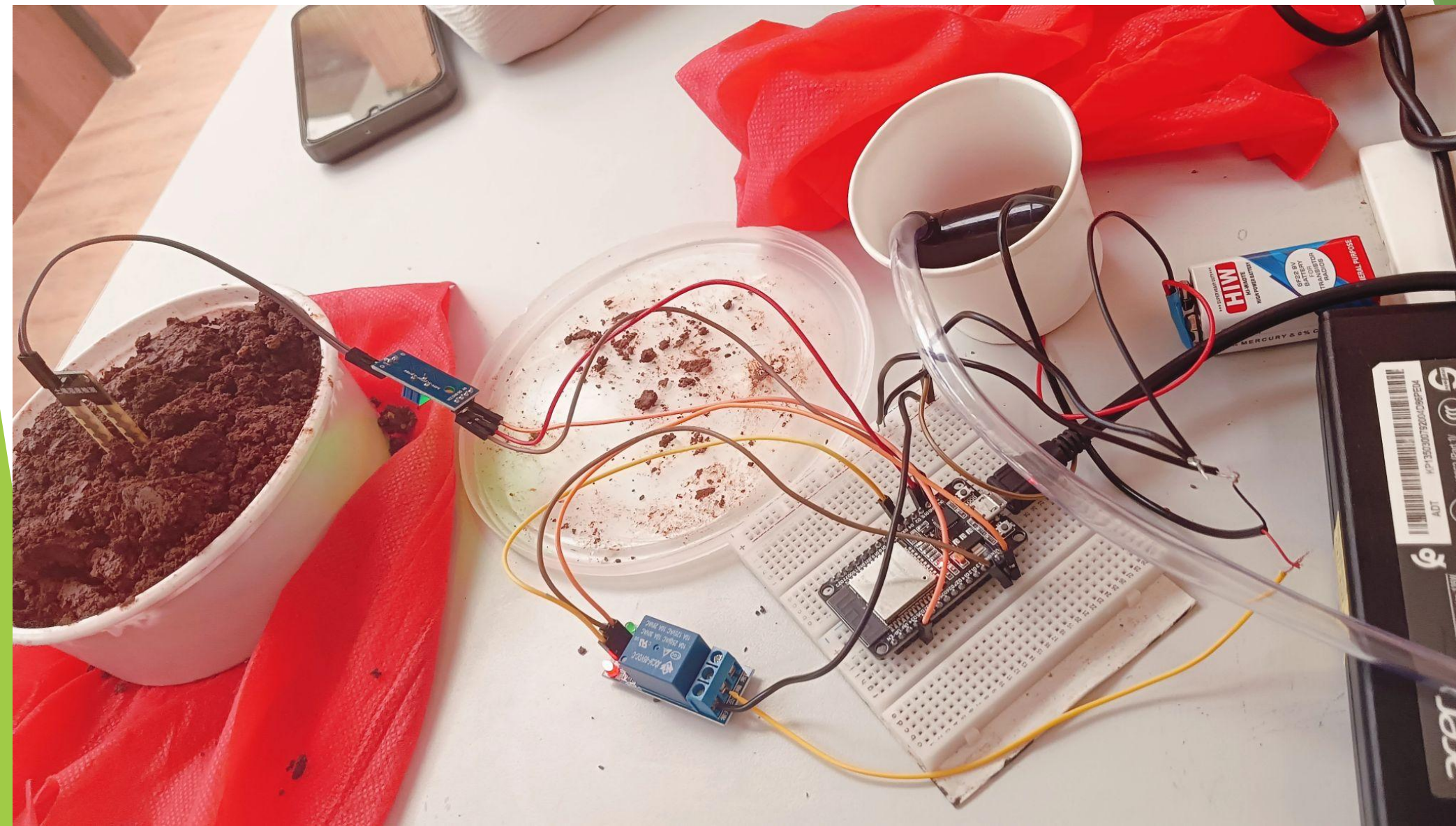
WORKING PRINCIPLE:

- ▶ The ESP32, being the main controller in the smart irrigation system, follows the principle of operation where several of the manual processes usually exercised by farmers or gardeners are replaced with automatic control based on collected and existing data. In wise irrigation systems, the ESP32 controller will receive data from moisture sensors powered on in the wheeled irrigators or the weather data is available to such sensors as humidity. This means that the ESP32 will've received input on checking whether the moisture content in the soil is optimal or not based on which the irrigation system can decide to turn off or turn on a water pump. Additionally. Furthermore, because of the Wi-Fi feature built into the ESP32, both the soil and the irrigation system can be controlled and monitored remotely with a mobile phone and website. The implication is a reliance on minimum water usage as opposed to the previous manual methods relying on an assumption of the dry soil condition.

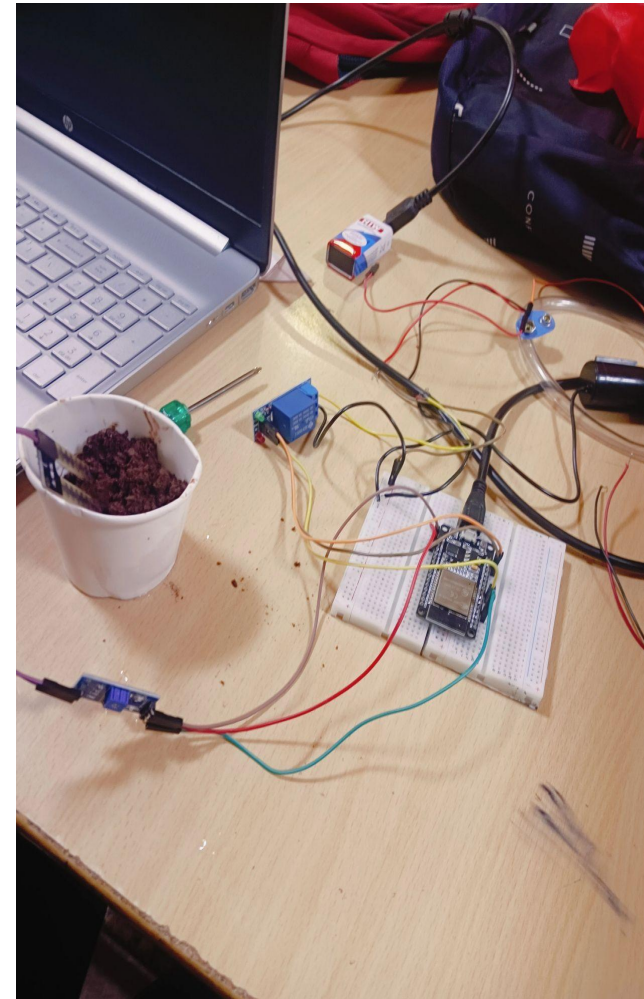
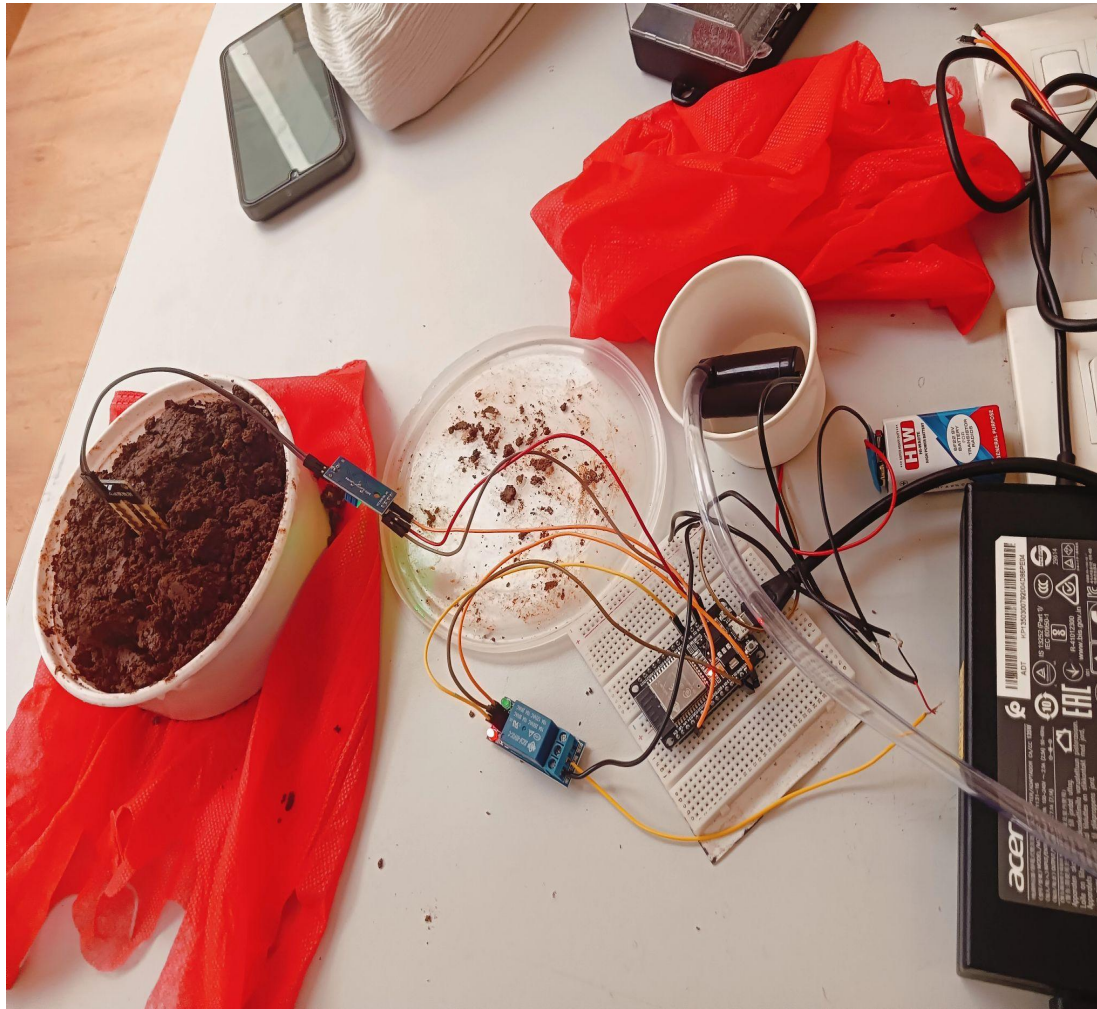
CIRCUIT DIAGRAM:



SNAPSHOTS OF THE PROJECT:



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OVERVIEW:

- ▶ A smart irrigation system using an ESP32 is an efficient and automated watering solution that leverages IoT technology to optimize water usage. It uses sensors to monitor soil moisture and environmental conditions in real time. The ESP32 processes this data and controls a water pump or valve to irrigate plants as needed. The system can integrate weather data to avoid overwatering during rainfall and supports remote monitoring and management through Wi-Fi connectivity. Users can access the system via a mobile app or web interface, ensuring precise control and reduced manual effort. This approach promotes sustainable irrigation by conserving water and adapting to environmental conditions.